

APB_DEVICE IP Specification

(RTL-Derived / Verification-Oriented)

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Purpose: This document reconstructs the functional specification that would normally accompany the RTL. It is intended as the primary reference for DV planning and interview preparation. Because the original design specification is unavailable, the RTL implementation is treated as the behavioral source of truth.

Specification status	RTL-derived; not the original vendor/spec document
Primary DUT	APB_DEVICE
Upstream interface	APB slave
Downstream interface	APB master

1. Scope and Functional Overview

APB_DEVICE is a control/DMA bridge that accepts APB transactions on its upstream interface and can either (a) access internal DMA control/status registers, (b) directly forward an APB transaction to a downstream APB slave, or (c) autonomously generate downstream APB transactions for DMA INIT and DMA COPY operations.

- Upstream side: APB slave behavior, including PREADY wait-state generation and PRDATA readback.
- Downstream side: APB master behavior, generating address, direction, select, enable, and write-data signals.
- DMA control: source address, destination address, transfer length, initialization value, copy sequencing.
- Interrupt/status: invalid operation, invalid length, overlap detection, DMA completion.
- Protocol checks: APB setup/access ordering and stability checks.

2. Architecture

High-level data/control flow:

Upstream APB Master -> APB_DEVICE -> Downstream APB Slave
|
+--> DMA control/status + interrupt

3. Interfaces

3.1 Clock / Reset

Signal	Dir.	Width	Description
pclk	I	1	DUT clock.
presetn	I	1	Active-low asynchronous reset.

3.2 Upstream APB Slave Interface

Signal	Dir.	Width	Description
paddr	I	32	APB address; paddr[31:28] selects function/register region.
pwrite	I	1	1=write, 0=read.
psel	I	1	APB slave select.
penable	I	1	APB access-phase indication.

pwdata	I	32	Write data.
prdata	O	32	Read data.
pready	O	1	Transfer completion; DUT may insert wait states.

3.3 Downstream APB Master Interface

Signal	Dir.	Width	Description
slave_paddr	O	32	Downstream APB address.
slave_pwrite	O	1	1=write, 0=read.
slave_psel	O	1	Downstream select.
slave_penable	O	1	Downstream access-phase indication.
slave_pwdata	O	32	Downstream write data.
slave_prdata	I	32	Downstream read data.
slave_pready	I	1	Downstream transfer completion.

3.4 Interrupt

Signal	Dir.	Width	Description
apb_device_int	O	1	Asserted when any bit of dma_int is set.

4. Address Map

The upper address nibble paddr[31:28] is the function selector. Lower address bits are forwarded for direct accesses; register regions are selected by the upper nibble.

paddr[31:28]	Symbol	Access	Function
0x0	DIRECT_ADDR	R/W	Direct downstream APB access.
0x1	DMA_SRC	R/W	DMA source address register.
0x2	DMA_DST	R/W	DMA destination address register.
0x3	DMA_LEN	R/W	DMA transfer length register.
0x4	DMA_INIT	W	Trigger/configure DMA initialization operation.
0x8	DMA_COPY	W	Trigger DMA copy operation.
0xF	DMA_INT	R/W	Interrupt status / clear register.
0x5-0x7, 0x9-0xE	Reserved/Invalid	-	Invalid operation; invalid_op event is generated.

5. Register Specification

Name	Width	Reset	Read	Write	Behavior
DMA_SRC	28	0	Yes	Yes	Program source address; auto-increments during DMA INIT/COPY downstream operation.
DMA_DST	28	0	Yes	Yes	Program destination

					address; auto-increments after completed DMA COPY write.
DMA_LEN	5	0	Yes	Yes	Transfer length; valid values 1..16. Acts as remaining-count during DMA.
DMA_INIT_VALUE	32	0	Internal	Internal	Captures DMA_INIT write data; also captures downstream read data during COPY.
DMA_INT	32	0	Yes	Yes	Sticky event/status bits. Write-one-to-clear behavior implemented.
PREADY delay	32	0	Internal	N/A	Randomized 0..9 cycle upstream wait-state counter used by DUT implementation.

5.1 DMA_INT Bit Definition

Bit	Name	Set condition
3	invalid_op	Upstream transaction selected no valid function region.
2	invalid_value_of_length	DMA_LEN write completes with value 0 or >16.
1	copy_overlap	DMA_COPY command detects overlapping source/destination ranges.
0	dma_init_copy_done	DMA INIT or COPY downstream transaction completes.

Interrupt status is sticky. A completed write to DMA_INT clears the selected bits using dma_int & ~pwwdata (W1C). The external interrupt is the reduction-OR of dma_int.

6. Functional Behavior

6.1 Direct APB Access

When paddr[31:28] == 0, the DUT treats the upstream transaction as a direct access and creates a corresponding downstream APB transaction. The downstream address uses paddr[27:0], while read/write direction and write data are forwarded from the upstream request.

- Direct read: downstream read is issued and returned slave_prdata is used as upstream prdata on completion.
- Direct write: downstream write is issued with the upstream write data.
- dma_direct_active tracks the outstanding downstream direct transaction.

6.2 DMA INIT

DMA INIT initializes a range starting from DMA_SRC using the data written to DMA_INIT. The operation becomes active after the upstream DMA_INIT transfer completes. For each downstream write completion, DMA_SRC increments and DMA_LEN decrements. The operation ends when the final transfer completes and generates the DMA completion event.

6.3 DMA COPY

DMA COPY transfers data from the configured source range to the destination range. The implementation uses two active-phase flags: P1 and P2.

- P1: drive a downstream APB read from DMA_SRC.
- When the read completes, the returned slave_prdata is captured and a downstream write toward DMA_DST is prepared.
- P2: complete the downstream APB write; after completion DMA_DST increments and DMA_LEN decrements.
- If more data remains, the state returns to P1 for the next element; otherwise the copy operation ends.

6.4 DMA Busy / Concurrent Access

The RTL derives an internal is_dma_busy condition from the active DMA phases and direct/terminal conditions. Upstream PREADY generation and direct-access admission use this busy status to limit conflicting operations while autonomous DMA activity is in progress.

6.5 DMA Copy Overlap

A DMA_COPY request generates the copy_overlap event when the source and destination ranges intersect under the implemented range comparison. This condition is reported through DMA_INT bit 1.

7. APB Timing and Wait-State Requirements

The DUT implements APB setup/access behavior on both sides. On the upstream side, PREADY may be delayed by a randomized 0..9-cycle counter. On the downstream side, the DUT waits for slave_pready before completing the current operation.

- A new APB transfer begins in setup phase: PSEL=1 and PENABLE=0.
- Access phase follows with PENABLE=1.
- When PREADY=0, the current transfer remains active; address/control/data must remain stable according to the implemented assertions.
- A transfer is considered complete when PSEL=1, PENABLE=1 and the relevant PREADY is asserted.

8. Reset Behavior

presetn is active-low asynchronous reset. The RTL resets control/status registers and operation state to inactive/default values.

State/Signal	Reset value
dma_src	0
dma_dst	0
dma_len	0
dma_init_value	0
dma_int	0
dma_direct_active	0
dma_init_active	0
dma_copy_active_p1	0
dma_copy_active_p2	0
downstream APB outputs	0 / idle

9. Error and Interrupt Behavior

Condition	Expected DUT behavior
Invalid address region	Generate invalid_op event; set DMA_INT bit 3.
DMA_LEN = 0	Generate invalid length event; set bit 2.
DMA_LEN > 16	Generate invalid length event; set bit 2.

Source/destination overlap on DMA_COPY	Generate overlap event; set bit 1.
DMA_INIT/COPY downstream completion	Generate done event; set bit 0.
Write DMA_INT with bit=1	Clear corresponding status bit (W1C).

10. Verification Requirements (DV View)

The following are the natural verification points implied by the RTL-derived specification. These are verification targets, not additional RTL behavior.

Feature	Key scenarios/checks
APB protocol	Setup/access ordering, wait states, stable address/control/data, back-to-back transfers.
Direct access	Read/write forwarding, downstream latency, read-data return, concurrent/busy behavior.
Registers	Reset values, read/write/readback, update only after transfer completion.
DMA_INIT	Length boundaries, sequential source addresses, repeated writes, wait states, done interrupt.
DMA_COPY	Read-then-write sequencing, data correctness, address increments, length decrement, final completion.
Error handling	Invalid address, illegal length, overlap boundary conditions.
Interrupt	Set, sticky behavior, W1C clearing, multiple-bit behavior.
Reset during activity	Reset while idle and while direct/DMA operation is active; outputs/state return to idle.

11. RTL-Derived Open Points / Clarifications

Because the original design specification is unavailable, the following items should be confirmed if a formal project specification is later obtained:

- Exact APB timing contract for upstream PREADY and downstream PSEL/PENABLE sequencing.
- Whether DMA_SRC/DST are byte addresses or word addresses; RTL stores 28-bit values and increments by 1.
- Exact semantics when an upstream request arrives while DMA is busy.
- Priority when a DMA interrupt event and a software W1C clear target the same bit in the same cycle.
- Exact behavior for DMA_LEN boundary values during active operations.
- Meaning and intended use of the internal MEM[0:1023] array; it is initialized in RTL but is not used in the functional datapath shown.
- The original intended role of the internal demonstration/self-check logic near the end of the RTL.

12. DV Quick Reference

For first-pass project understanding, the DUT can be remembered as:

APB Slave (configure/control) -> DMA/Direct Engine -> APB Master (execute access)

Registers: SRC / DST / LEN / INT

Commands: INIT / COPY

Events: invalid op / invalid len / overlap / done

State: direct / INIT / COPY-P1 / COPY-P2